



**Multi-Task Learning for Sculpture Classification:
Focusing on Materials, Genres, and Years**

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Abstract

Due to the complexity of sculpture materials and genres, the classification and digital management of sculptures are two major challenges faced by museums and heritage organisations. Traditional methods of management and classification rely heavily on manual inspection, which not only reduces efficiency but also consumes significant labour costs. Therefore, this research aims to develop a multi-task learning model that automates and enhances this classification process helping museums and heritage organisations to address these challenges. The research questions of this paper focus on the performance of classifying the materials, genres and years of sculptures using a multi-task learning model and investigate whether the tasks of multi-task learning benefit from learning shared representations compared to single-task learning.

In this paper, a multi-task learning approach is applied to extract shared features from sculpture images using a pre-trained ResNet50 as a base. Layers applied to specific tasks are then used to classify sculptures into multiple categories of materials, genres, and years. We used cross-entropy loss for each task and balanced the learning process using dynamic weight averaging (DWA). The training results of the model show that multi-task learning performs best on the genres of the sculptures and worst on the years. Overall, all three tasks of multi-task learning performed better than single-task learning, suggesting that the tasks do benefit from sharing representations between them. The case of model classification accuracy is also related to the balance of the dataset and the visual features. This study fills the gap in applying multi-task learning for classification in the field of materials and sculpture, and lays the foundation for future research. On the other hand, the model provides an organisational framework for museums and heritage institutions to support the management and classification of sculptures, which will greatly improve efficiency and reduce labour costs.

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1. Introduction

1.1 Motivation

1.1.1 The Evolution and Challenges of Sculpture

Sculpture is a very common cultural heritage and art category. It shapes materials into works of art with a sense of space and texture by carving, moulding, combining and assembling. There is a long history of sculpture, dating back to the days of ancient civilisations. From the pyramids of Ancient Egypt to Greco-Roman statues, to the Renaissance statue of David, sculpture has always played an important role in cultures. It is one of the art forms in which mankind creates images capable of three-dimensional representation. The modern sculpture has also continued to evolve and innovate. Artists have begun to experiment with new materials and techniques, such as steel, plastics, and electronics, to create even more original and unique works.

Museums and heritage organisations currently face challenges in managing and categorising sculpture. On the one hand, sculptures are made from a wide range of materials, each with unique physical and chemical properties, which complicates the material classification and digital archiving of sculptures. This complexity is exacerbated by sculptures made from a mixture of materials. Traditional methods of classifying sculptures rely heavily on manual inspection, which is not only time-consuming and inefficient, but also prone to human error. On the other hand, identifying the artistic style and historical period of a sculpture requires extensive expertise and experience, which can be costly. In the study of Lithuanian Municipal Museums, it was noted that museums face a lack of skilled and professional staff and excessive workload when implementing digital management (Kimura, 2022). This suggests that museums lack the necessary resources and organisational framework to support management and classification efforts. Therefore, the development of a model to automate this range of classification processes may help museums and arts organisations to address these issues.

1.1.2 The Value of Deep Learning in Sculpture Classification

Deep learning, as one of the mainstream techniques of artificial intelligence, can classify sculptures efficiently and effectively. Deep learning processes and analyses large amounts of data by simulating the structure and function of the human brain through neural network models. This technology is now widely used in tasks such as image recognition, natural language processing, and speech recognition. With the continuous upgrading of technology and the expansion of application of long techniques, deep learning technology will play an even more important role in the cultural heritage field and museums in the future. It will gradually become an important tool for the protection, restoration and research of art and cultural heritage. Multi-task learning is an extension of deep learning. By learning multiple related tasks at the same time, this approach can not only improve generalization performance but also save storage space and cut down on the time it takes to make prediction (Zhang and Yang, 2021). This advanced technique is better able to achieve simultaneous classification of multiple features of the sculpture and improve the accuracy of the model classification.

Therefore, the motivation of this study is to create a sculpture classification model through a multi-task learning approach that will streamline the current costly, inefficient classification process in museums and art organizations. Thereby addressing the current shortcomings and significantly improving the management and preservation of cultural heritage.

1.2 Research Questions

This research will create a sculpture classification model using multi-task learning. Based on the pictures of the sculptures, the sculptures are classified in terms of material, genre and year of creation. By automating the classification process, the model reduces the reliance on manual labour, minimises errors and accelerates the digitisation of museum collections. Here are descriptions of the three tasks in multi-tasking learning.

Materials Classification: This task focuses on accurately identifying and classifying different materials based on their features.

Genres Classification: This task involves classifying genres, which enhances the overall classification process by improving the feature extraction.

Years Classification: This task aims at classifying years, providing temporal context that complements the other tasks, helping to understand how the characteristics of materials and genres might be influenced by different historical periods.

Based on these three classification tasks, the following research questions were posed:

1. How does multi-task learning perform in classifying the materials, genres and years of creation of sculptures? Are there significant differences in performance between these tasks?
2. Did multi-task learning improve the accuracy of the materials, genres and years of creation of sculptures compared to the single-task learning model? In which tasks is the improvement more significant?
3. How do the characteristics of the dataset such as category distribution, and visual features affect the classification of the multi-task learning model?

1.3 Structure

This study will consist of five main parts. The first section (Literature Review) will discuss the previous articles and studies related to the topic of this research, summarising the current research gaps. The second section (Methodology) will detail the methodology used to achieve the research objectives. It is divided into dataset management and model construction. The third section (Results) will demonstrate the classification accuracy and performance metrics of the model. The fourth section (Discussion) will explain the significance and implications of the results. The final section (Conclusion) will summarise the main findings of the study, the limitations of the study and indicate future research directions.

2.Literature Review

2.1 Section Introduction

This section will review the current literature related to the research topics, starting with a broad overview and progressively focusing on more specific details. It will first introduce the broad topics of machine learning in cultural heritage and art, followed by the historical and artistic context of sculptural materials. Next, the application of machine learning in material classification will be further discussed. Multi-task learning as an advanced machine learning technique will be explained in more depth and its application to art classification will be discussed. Finally, the status and significance of this research is summarised.

2.2 Machine Learning in Cultural Heritage and Art

Machine Learning (ML) has become crucial in the field of cultural heritage and art. In the field of cultural heritage, it provides innovative solutions for accessibility, preservation and analysis. One significant application is the automated recognition and classification of monuments using Convolutional Neural Networks (CNNs), which helps in preserving and promoting cultural heritage by enabling users to access detailed information about historical sites through mobile applications (Kharghar et al., 2023). For the conservation of cultural heritage buildings, deep learning methods like transfer learning with pre-trained networks and Grad-CAM are used to identify and localize defects, facilitating timely and accurate renovation processes (Bahrami and Albadvi, 2023). The identification and dating of heritage objects have also benefited from ML, with near-infrared spectroscopy combined with supervised learning methods providing highly accurate dating of paper books (Coppola et al., 2023).

Machine learning has a wide range of applications in the art field as well. One of the important applications is the automatic analysis and classification of art images. Where machine learning techniques such as residual networks help to classify paintings and identify features for image retrieval in online art collections (Kvak, n.d.). Moreover, machine learning has been used to authenticate artefacts, for example the use of residual neural networks on the Dutch National Museum dataset significantly improved

the accuracy and reliability of the artefact authentication process (Dobbs and Ras, 2022). Finally, machine learning-based tools can analyse user behaviour in arts-related online searches, providing insights that can help cultural institutions enhance their digital presence and better understand user preferences (Pergantis et al., 2023).

Despite these theoretical advances, challenges remain at the practical application level. For example, the application of AI and machine learning in Swedish cultural heritage institutions is at a very early stage. Key challenges include a lack of institutional and professional expertise, specific digitisation needs of collections, and resource constraints (Griffin et al., 2023). In addition, the application of machine learning in cultural heritage and the arts must consider socio-technical aspects. Detailed guidelines and checklists are needed to ensure responsible and ethical use of machine learning technologies (Lee, 2022).

In summary, machine learning applications in cultural heritage and art underscore its transformative impact in preservation and promotion, despite practical challenges.

2.3 Historical and Artistic Context of Sculpture Materials

The materials used in sculpture significantly influence our understanding of the period and genre, reflecting the cultural and historical contexts. Historically, materials like clay, wood, ivory, stone, and metal were selected based on their availability and the technological capabilities of the time, as seen in ancient Greek and Roman sculptures where marble was highly prized for its workability and aesthetic qualities (Yavuz, 2021). In contrast, the Iron Age societies in the southeastern Iberian Peninsula used locally sourced limestone for their zoomorphic sculptures, reflecting both the material's availability and the cultural significance of the genre (Li and Li, n.d.). The Industrial Revolution also significantly influenced the choice of sculpture materials by expanding the range of available materials and introducing new techniques. Initially, sculptors like L. R. Kitchin, who began with classical materials and styles, shifted to using industrial materials such as metal and welding equipment (Kitchin and Brown, 1979). The evolution of materials over time, from terracotta in ancient Crete to modern materials like bronze and welded steel, shows a continuous adaptation to new artistic

expressions and technological advancements (Lazzarini and Marconi, 2014) (Greene-Mercier, 1982). The genre of the sculpture also dictates material choice; for instance, religious sculptures often used materials that conveyed grandeur and permanence, such as the polychrome wooden sculpture of Saint Dominic from the 18th century, which utilized materials like gypsum, gold leaf, and various pigments to achieve its detailed and expressive finish (Knappett and Poursat, 2022). Humanistic forms often utilize materials that evoke warmth and familiarity, such as bronze or softer metals, to resonate with human experiences and emotions (Sinclair, 1967).

However, multi-material sculpture can pose challenges to understanding and the historical and artistic context of sculpture. For example, the restoration of polychrome wooden statues, such as the one revered as St. John in Sardinia, involves identifying various materials like pigments and binders, which have been modified over time to meet changing artistic and religious needs (Lo Monaco et al., 2022). A sculpture may use both old stone and newer materials, which can present a challenge to art historians as they need to determine where each material came from and when it was added.

Thus, the interplay between period and genre significantly shapes the material choices in sculpture, reflecting broader historical, cultural, and technological trends.

2.4 Machine Learning for Material Classification

Machine learning for material classification has made significant advances, especially in situations where different and visually similar materials need to be accurately recognised. Various machine learning techniques such as Support Vector Machines (SVMs), Random Forests and Neural Networks are compared for waste classification. These techniques demonstrated their ability to handle different material types such as metals, plastics and organic materials, improving the efficiency of the classification task (Rashidi et al., 2015). In addition, combining image processing techniques with machine learning models can be effective in dealing with differences in material appearance, such as between various types of plastics, metals and organic waste. This approach uses pre-processing steps such as image enhancement and segmentation to improve the quality of the input data prior to classification, thereby

improving the performance of the machine learning model (Satvilkar, 2018). Deep learning, specifically Convolutional Neural Networks (CNNs), have been shown to be very effective in recognising and classifying materials from images. Achieving high accuracy in material classification through CNNs requires the use of well-annotated datasets to address the challenges associated with the texture and appearance of materials (Penumuru et al., 2019).

Advanced neural network architectures such as PSPNet can facilitate accurate material classification even when materials exhibit visually similar characteristics. Accurate annotations play a crucial role in this process, ensuring that machine learning models can effectively distinguish between different material types, as well as subtle differences between similar materials (Zhang and Li, 2022). Finally, the combination of supervised learning with a knowledge base for automatic content enrichment of images shows the potential for enhancing image metadata with semantic information. This approach utilises classifiers such as SVMs and other machine learning algorithms to identify materials in an image, for example identifying specific types of building materials, fabrics and composites, which are then enriched with relevant metadata. Integration of a knowledge base automatically correlates additional contextual information, making the classification process more informative and comprehensive (Mallepudi et al., 2011).

In summary, machine learning techniques, including traditional algorithms and advanced deep learning models, significantly improve the accuracy and efficiency of Material Classification. However, most of the material classification are single-task learning methods. Research on Multi-Task Learning methods applied to material classification needs to be further strengthened.

2.5 Multi-Task Learning

Multi-Task Learning (MTL) is a more advanced machine learning technique and is also seen as an important extension and enhancement to traditional single-task learning methods. MTL significantly impacts the generalization ability of deep learning models by leveraging shared knowledge across multiple tasks, enhancing performance and

robustness. MTL improves generalization by maximizing positive transfer between tasks while reducing task interference, as seen in the Multi-Task Structural Learning (MTSL) approach, which dynamically learns both the architecture and parameters of the model (Gurulingan et al., 2023). MTL frameworks that jointly learn latent features and explicit task relations, such as modelling task relations through input gradients, have shown to reduce generalization error (Bai and Zhao, 2022). The hard parameter sharing approach in MTL creates an inductive bias that can improve generalization and reduce computational costs, although it requires careful gradient management to avoid task conflicts (Meshgi et al., 2022). Semi-supervised MTL models like MultiMix, which jointly learn disease classification and anatomical segmentation, demonstrate improved generalizability in medical imaging tasks (Nakamura et al., 2022).

However, MTL also brings some challenges. Despite these benefits, MTL can sometimes hurt generalization due to spurious correlations, where non-causal knowledge from one task negatively impacts another. Addressing this, several advanced architectures have been proposed. The Multi-gate Mixture-of-Experts (MMoE) model consists of shared experts and task-specific gating networks. Each gating network determines the contribution of each expert to the task, enabling the model to dynamically balance task-specific and shared knowledge (Ma et al., 2018). Furthermore, the Mixture-of-Experts (MoE) architecture in multi-task learning benefits from expert specialisation but requires effective gating mechanisms. Existing gates like Top-k are not smooth, causing convergence issues. DSelect-k introduces a differentiable and sparse gate to the MoE architecture. It provides a smooth and differentiable alternative that improves training efficiency and prediction performance (Hazimeh et al., n.d.). Another novel architecture is Cross-Stitch Networks. It introduces a novel unit that combines the activation of multiple networks to learn shared representations in a multi-task learning setup. This facilitates the model to dynamically balance shared features with task-specific features, thus improving cross-task performance (Misra et al., 2016). Finally, Multi-Task Attention Network (MTAN) uses an attention mask to determine the importance of shared features for each task, balancing task sharing and task-specific learning (Liu et al., 2019a).

Overall, MTL's ability to share knowledge across tasks, when managed effectively, can significantly enhance the generalization ability of deep learning models. The multi-

tasking learning library LibMTL (Lin and Zhang, 2022) used in this study supports the advanced architecture mentioned above.

2.6 Multi-Task Learning for Art Classification

Multi-task learning (MTL) has proven to be highly effective in classifying art by leveraging shared representations across multiple related tasks. For instance, ArtSAGENet integrates Graph Neural Networks (GNNs) and Convolutional Neural Networks (CNNs) to jointly learn visual and semantic-based artistic representations, outperforming single-task alternatives in tasks like style classification, artist attribution, and creation period estimation (Efthymiou et al., 2021). Similarly, a deep multi-branch and multitask neural network exploits different resolutions of painting crops to categorize artist, style, and genre, achieving significant accuracy on the MultitaskPainting100k dataset (Bianco et al., 2019). The use of knowledge graphs combined with content analysis enhances classification performance by modelling relationships between labelled and unlabelled data, as demonstrated in the classification of paintings and Buddha statues (Sabatelli et al., 2019). Additionally, the OmniArt model uses a shared representation layer to simultaneously predict multiple attributes of an artwork, such as artist, period of creation, and material. This is achieved through a basic feature extraction layer and a shared representation layer, which then branches into separate task-specific layers (Strezoski and Worring, 2017). Sato et al. applied MTL to analyse the artistic differences between different artistic domains such as painting, sculpture, architecture and gardening. Their model also uses a shared layer to capture the common features of different tasks, which helps to understand the artistry involved in each domain (Sato et al., 2022).

Overall, MTL's ability to integrate various data sources and learning tasks makes it a powerful approach for the comprehensive classification of art. However, in using MTL to classify art, most studies have focused on painting as an art form, for example, classifying the artists, genres and years of painting. Or they focus on the broader concept of art, classifying painting, sculpture and architecture. But there is very little research that focused on sculpture, especially in terms of its materials, genres, and years.

2.7 Summary

Machine learning in cultural heritage and art show its strong potential to improve preservation and promotion, even though there are practical challenges. Sculpture can be both a cultural heritage and an art form. Its materials are influenced by historical periods and artistic genres, reflecting the cultural and technological contexts of the time. Even though machine learning as a technique has many good research results on classification of materials. However, most of the studies are using single-task learning approach and lack of multi-task learning application in this area.

Multi-Task Learning (MTL) can enhance the generalization ability of deep learning models by leveraging shared knowledge across multiple tasks, thus improving performance and robustness despite some challenges like task interference. Applying MTL to classification has proven highly efficient, significantly improving the accuracy of models by integrating shared representations across tasks, which validates its use in classifying sculpture materials, genres, and creation times in this project.

While MTL has been successfully applied to classification tasks, such as identifying painting styles, artists, and creation periods, there is a significant gap in its application to sculpture, particularly for material classification. This gap highlights the need for more focused research in using MTL to classify sculpture, an area currently underexplored.

3. Methodology

This chapter consists of two sections, Dataset Management and Model Construction.

3.1 Dataset Management

This section details the process of managing the sculpture dataset from the source of the data to the preparation of the data. The dataset consists of images of sculptures and metadata such as materials, genres, and years. After the raw data was cleaned and transformed, cluster analysis was used to simplify the categories. Then label encoding was performed on the categories of all three variables. The final dataset has a total of 3679 samples, which were divided into training set, validation set, and test set to ensure that they can be used in deep learning models.

3.1.1 Data source

The data source for this project is the Art Database under Wikidata ("Wikidata," n.d.), which covers more than 1.7 million art-related data. These data are collated from publicly available images from various museums, galleries, and art collections around the world. More than seventeen thousand pieces of sculpture-related data have been extracted from it. They are mainly images of sculptures and the corresponding metadata, including materials, genres, and years. Materials and genres are multi-label categorical variables, and years is numeric variable. Below is a table of variable descriptions for the raw data (Table 1).

Table 1. Variable description table for raw data

Variables	Description	Examples
materials	Material used for sculpture	"marble, granite, bronze", "steel"
genres	Type of sculpture	"public art, equestrian statue", "public art"
years	Year of creation	"1959-01-01T00:00:00Z", "1986-01-01T00:00:00Z"

3.1.2 Data cleaning

Missing value cleaning was performed first. Since all three variables (materials, genres, years) have a large number of missing values, as shown in Figure 1, materials has 56% of the missing values, while genres and years have 63% and 44% of the missing values, respectively. This will have a negative impact on the training of the model. For all three tasks to be trained fairly, the missing values for the materials, genres and years variables were removed. This ensured that the metadata corresponding to the images were all complete, and left room for later processing by converting the few categories of materials, genres to the ambiguous category of 'other'. If we choose to keep the missing values as 'unknown', it will introduce one more ambiguous category, which will make the training of the model more difficult.

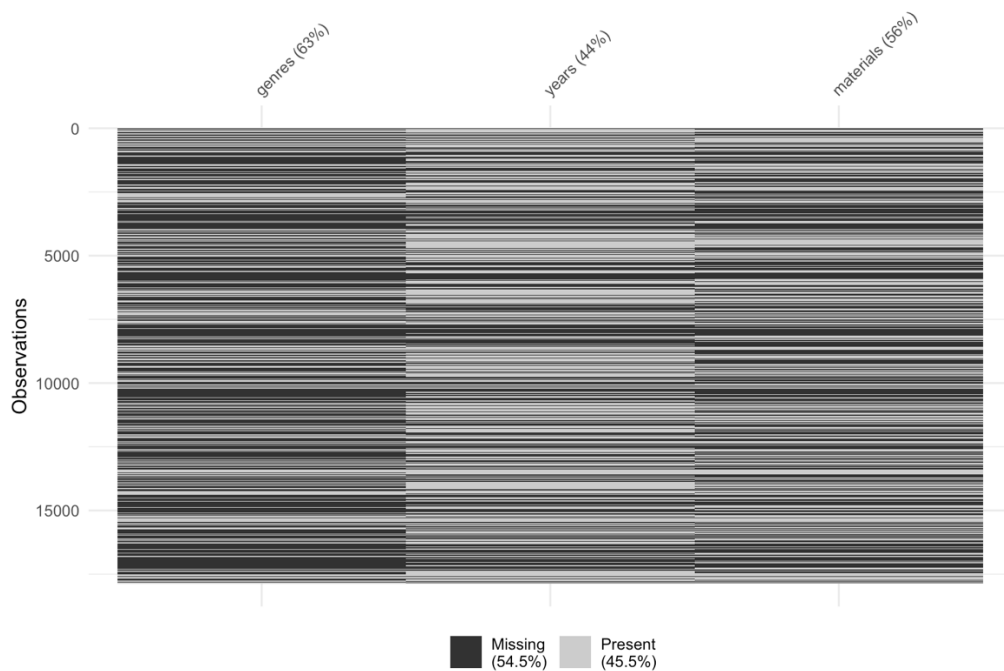


Figure 1. Missing Data for Sculpture Dataset

3.1.3 Data Wrangling

Data Wrangling is performed next. for years, as shown in table 1, the format of the raw data is very complex, so only years are extracted to simplify the format. Some of the materials and genres in the sculpture samples in the raw data have only one category, while others belong to several different categories at the same time, which means that they are multi-label categorical variables. By manually examining the images of the

sculptures corresponding to these multi-label categorical variables, we found that the first label in the multi-label category is often the most dominant or representative material or genre in the sculpture. For example, when one of the samples has the category of materials as 'bronze, stone', the sculpture in the image has a large area of bronze, and only a small portion of stone. This suggests that the first label is the dominant material in the image. This observation is consistent across the rest of the sample. To simplify the model difficulty, the secondary labels in the multi-label categories of materials and genres were removed, and only the first label in the multi-label category was retained. This ensured that there was only one primary label for all the variables in the category, which means that a sample only corresponded to a unique material and genre. In summary, simplifying multi-label categories into single-label categories avoids the complexity of multi-label classification in multi-task learning and lays the groundwork for subsequent multiclass classification.

3.1.4 Data distribution and K-means clustering:

For the two categorical variables materials and genres, two problems were identified by calculating and visualising the distribution plots (Figure2 & Figure3).

The first problem is that both categorical variables have too many categories and there is semantic overlap between several categories. Materials variable has 200 categories and genres has 102 categories, which is extremely challenging for the model to discriminate between categories. To solve this problem, the number of categories needs to be reduced.

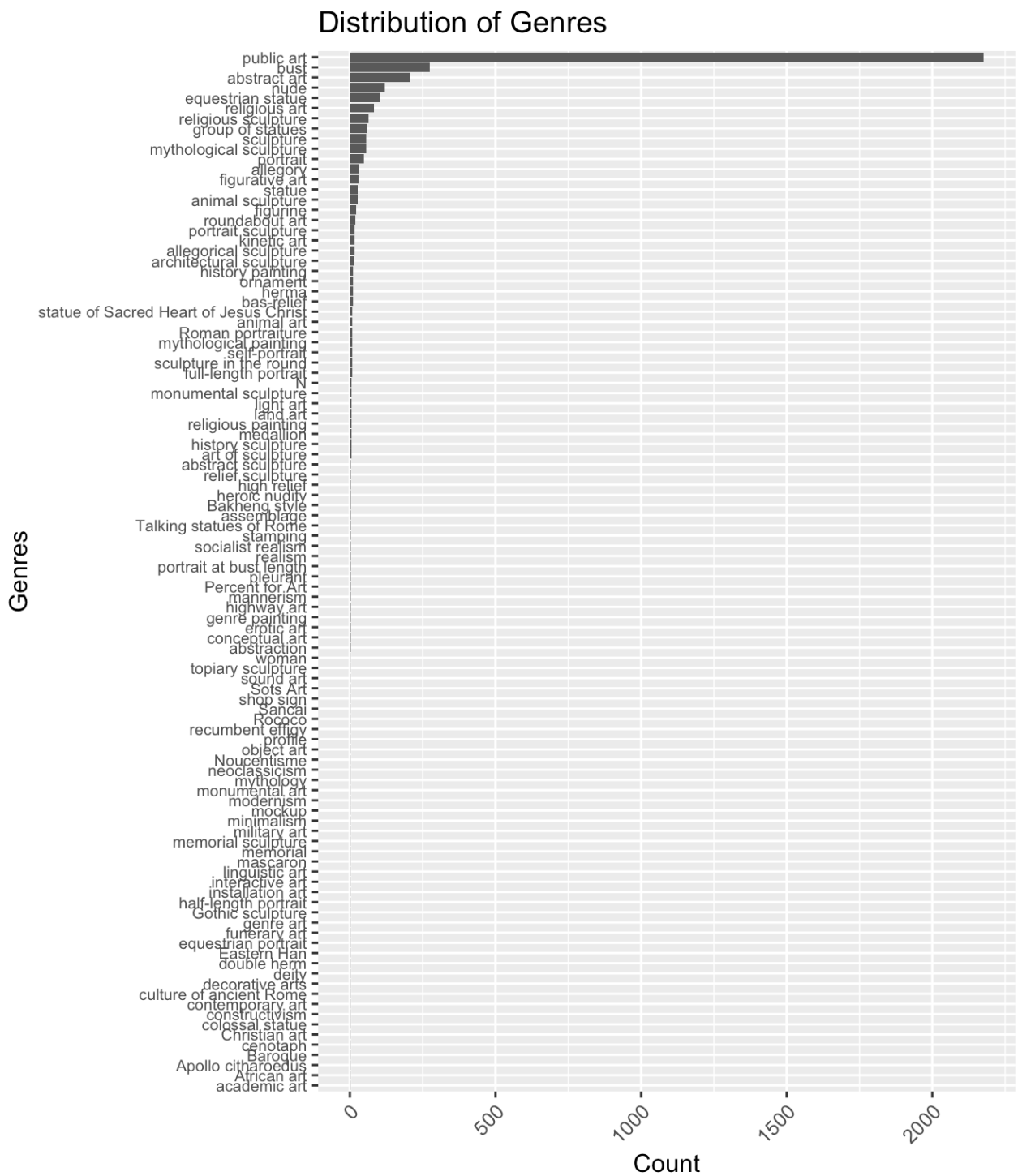


Figure 2. Distribution of Genres

Distribution of Materials

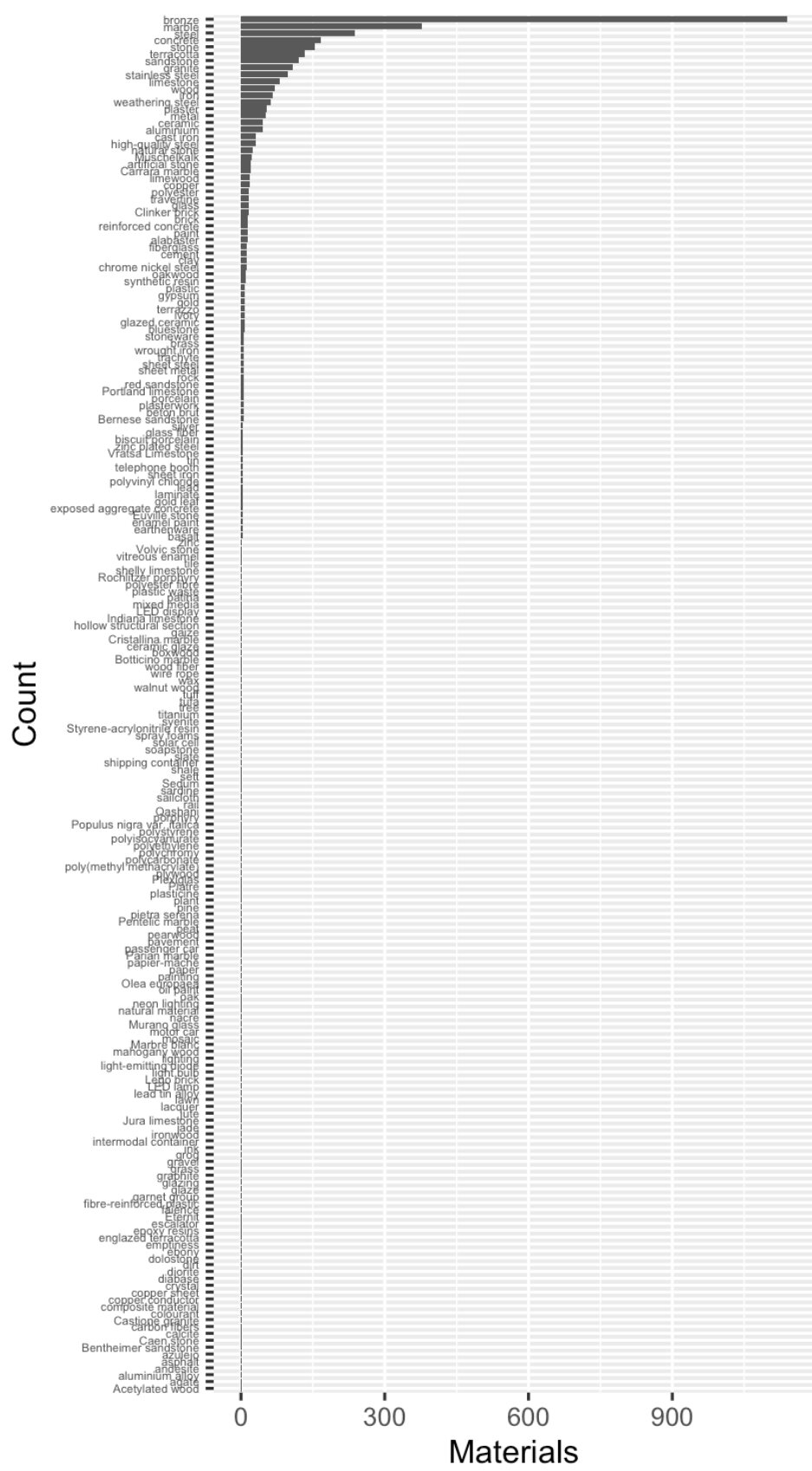


Figure 3. Distribution of Materials

A further look at the names of these categories revealed that there were nested relationships between the categories. In order to further explore these nested relationships, k-means clustering was performed on the category names of both of 200 materials and 109 genres using the Sentence Transformer model (Reimers and Gurevych, 2019). Sentence Transformer is a deep learning model for encoding sentences and computing semantic similarity between sentences. It works by converting text data into numerical embeddings and then applying k-means algorithm for clustering based on these embeddings. In this way, semantically similar texts will be grouped into the same cluster.

The optimal number of clusters can be determined by the silhouette score, with a higher silhouette score indicating a greater degree of cluster separation. From Figure 4 and Figure 5, the highest and most reasonable silhouette score for the materials clustering is when the number of clusters is 8, while the genres clustering reaches its highest score when there are 11 clusters. This indicates that clustering is most effective when the categories of materials and genres are divided into 8 and 11 clusters. The results of clustering (Figure 6 & Figure 7) reveal the semantic similarity between different materials as well as different genres. By visualising the clustering diagram and labelling the categories that occur most frequently in that clustering, it is possible to identify cases where one category is included in another. Combining the clustering results with personal knowledge and experience, highly relevant categories were merged. For example, the categories of materials such as 'oakwood', 'boxwood', 'walnut wood', 'pine', 'mahogany wood', 'chestnut wood', etc. have been merged into the category 'wood'. The genres 'abstract art', 'abstract sculpture', 'abstraction' has been merged into the category 'abstract art'.

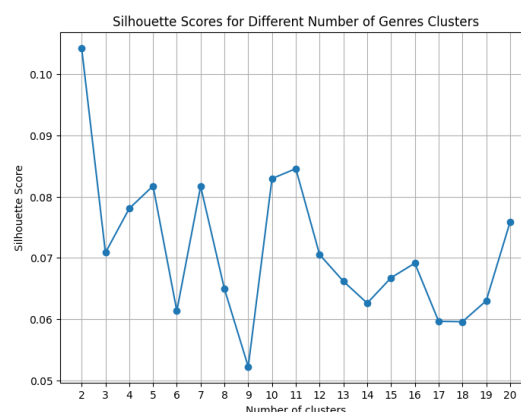
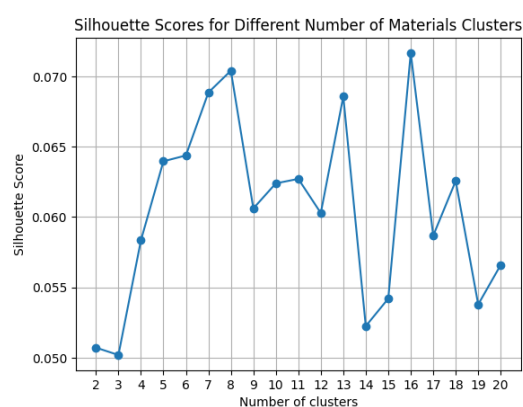


Figure 4. Silhouette Scores of Materials Clusters Figure 5. Silhouette Scores of Genres Clusters

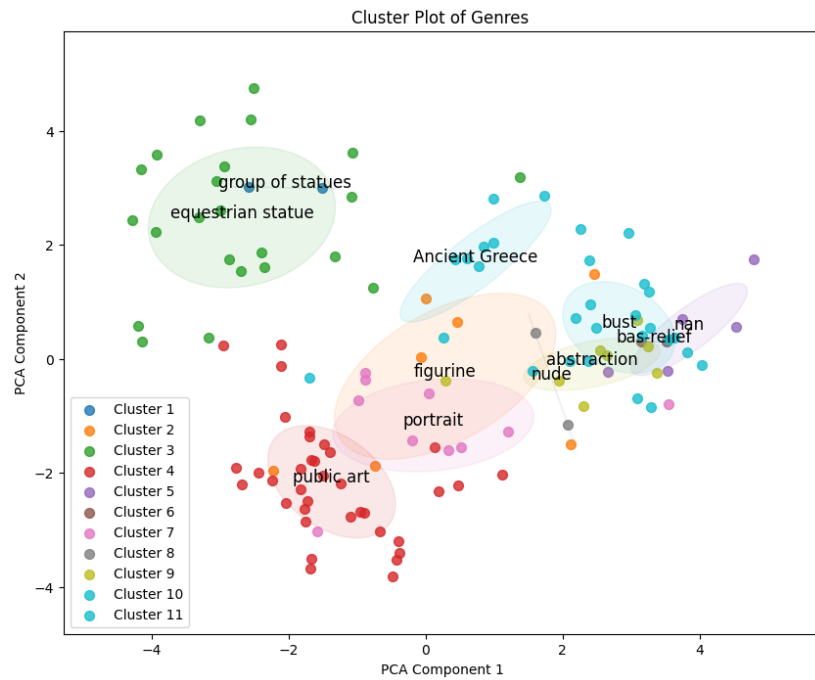


Figure 6. Cluster plot of genres

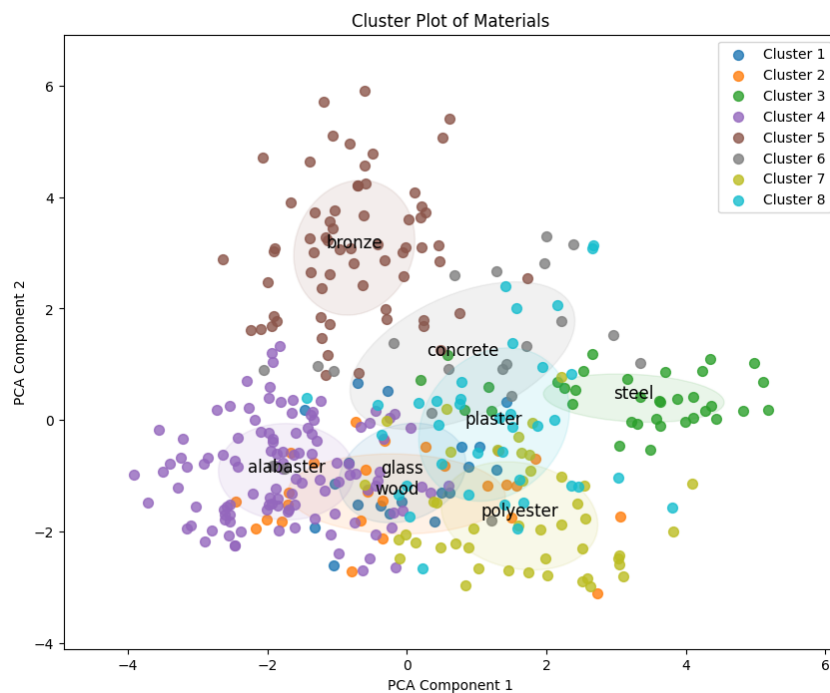


Figure 7. Cluster plot of materials

The second problem is the extreme infrequency of some categories. As can be seen from the distribution plots of these two variables (Figure 2 & Figure 3), they are both very unevenly distributed and there are many rare categories with too few frequencies. Even though the first problem was solved by combining as many categories as

possible in proximity, several categories still had very small sample sizes. To solve this problem, the rare categories with less than 1% of the total data were defined as 'Other'. The data distribution of the simplified materials and genres was visualised in Figure 8 and Figure 9.

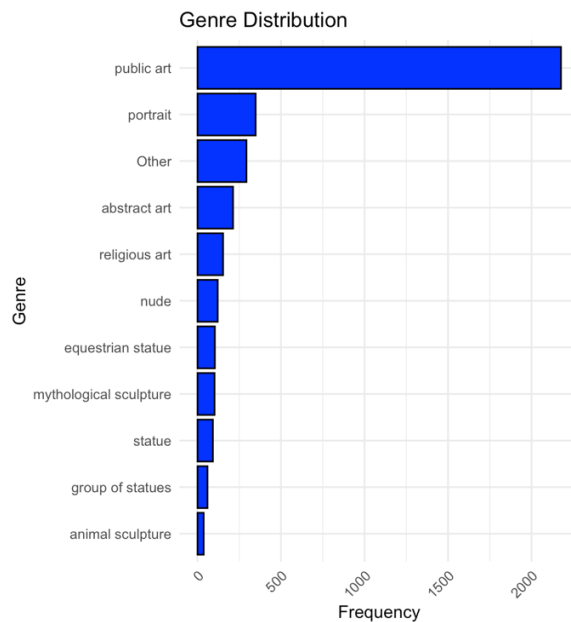


Figure 8. Genres distribution

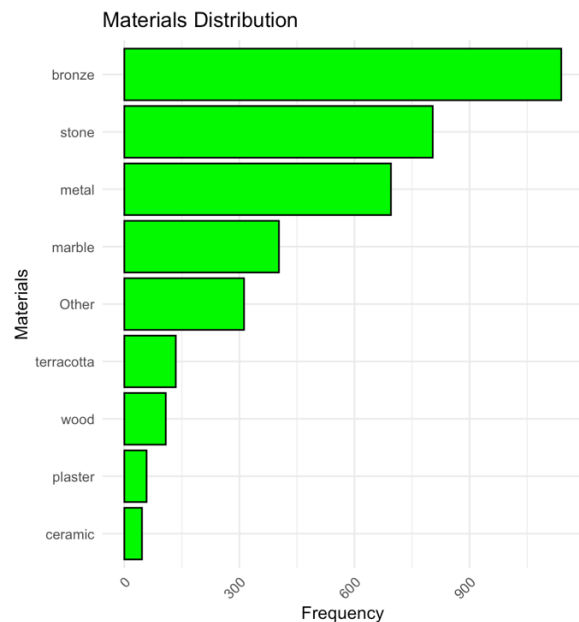


Figure 9. Materials distribution

For the numeric variable years, the maximum value is 2500 and the minimum value is -7500. About 98% of the years data are concentrated in the AD and only 2% are in the BC, which suggests that the data are concentrated in the years after the AD. There are also years of 2500 in the dataset, which is unrealistic data. Based on the above analysis, only data with years greater than 0 and less than or equal to 2022 were retained. The violin plot (Figure 10) shows that the distribution of the years variable is still very unbalanced after the initial processing. Therefore, it was binned and converted into a categorical variable. If Equal-frequency binning is used, the distribution will reach a balanced state, but the span of years for some of the bins will be extreme. Thus, the pattern of binning was customised (Figure 11) to achieve a balance between the bin's frequency and the bin's width.

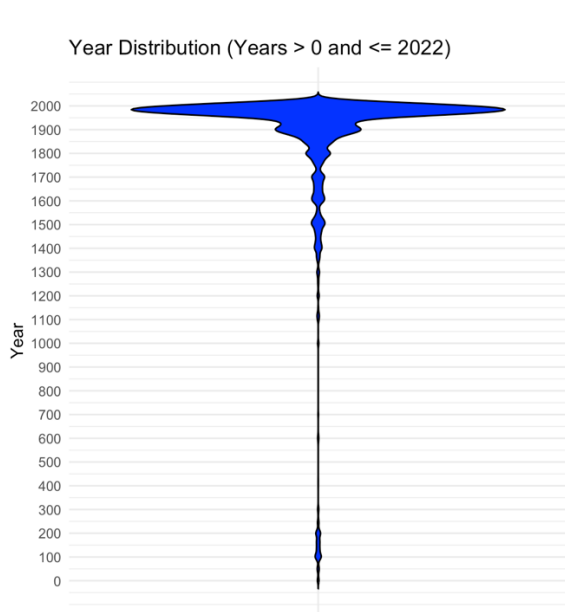


Figure 10. Years distribution

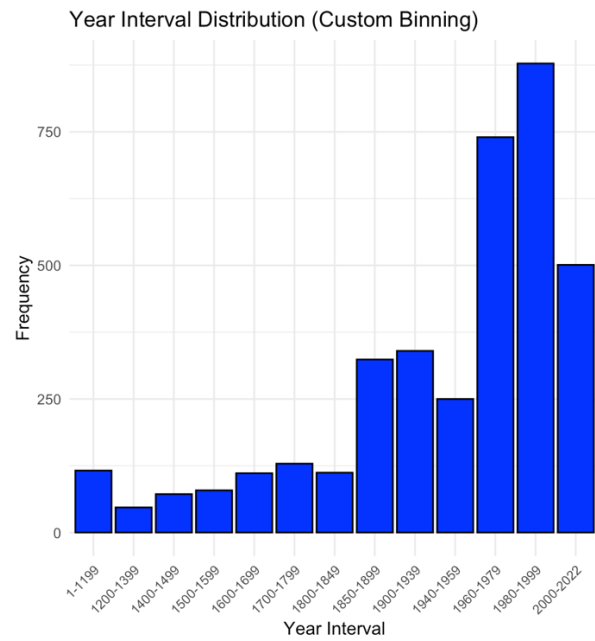


Figure 11. Year interval distribution

3.1.5 Final dataset

The categorical variables materials, genres and years are converted into numerical form by label decoding them. Each category in the variables was assigned to a unique integer, which helps in the input of the deep learning model. Then the dataset was randomly divided into three parts: 70% for training, 15% for validation and 15% for testing. The final dataset was obtained by processing the data as described above. The dataset has a total of 3679 samples and three categorical variables, of which materials has 9 categories, genres have 11 categories, and years has 13 categories. The number of each category is shown in Table 1, Table 2 and Table 3.

Table 2.Materials distribution Table 3.Genres distribution Table 4.Years distribution

Materials Distribution

	Count	Percentage (%)
bronze	1139	30.792106
stone	804	21.735604
metal	695	18.788862
marble	403	10.894836
Other	312	8.434712
terracotta	134	3.622601
wood	108	2.919708
plaster	58	1.567991
ceramic	46	1.243579

Genres Distribution

	Count	Percentage (%)
public art	2176	58.826710
portrait	348	9.407948
Other	293	7.921060
abstract art	213	5.758313
religious art	153	4.136253
nude	121	3.271154
equestrian statue	104	2.811571
mythological sculpture	102	2.757502
statue	92	2.487159
group of statues	60	1.622060
animal sculpture	37	1.000270

Years Distribution

	Count	Percentage (%)
1980-1999	878	23.736145
1960-1979	740	20.005407
2000-2022	501	13.544201
1900-1939	340	9.191673
1850-1899	324	8.759124
1940-1959	250	6.758583
1700-1799	129	3.487429
1-1199	116	3.135983
1800-1849	112	3.027845
1600-1699	111	3.000811
1500-1599	79	2.135712
1400-1499	72	1.946472
1200-1399	47	1.270614

3.2 Model Construction

In this section, we explain how the multi-task learning model is constructed to classify the sculpted images into multiple categories on materials, genres, and years, respectively. This section firstly shows the overall structure of the multi-task learning model in this study, then analyses the key layers in the model one by one, and finally presents the details of implementation.

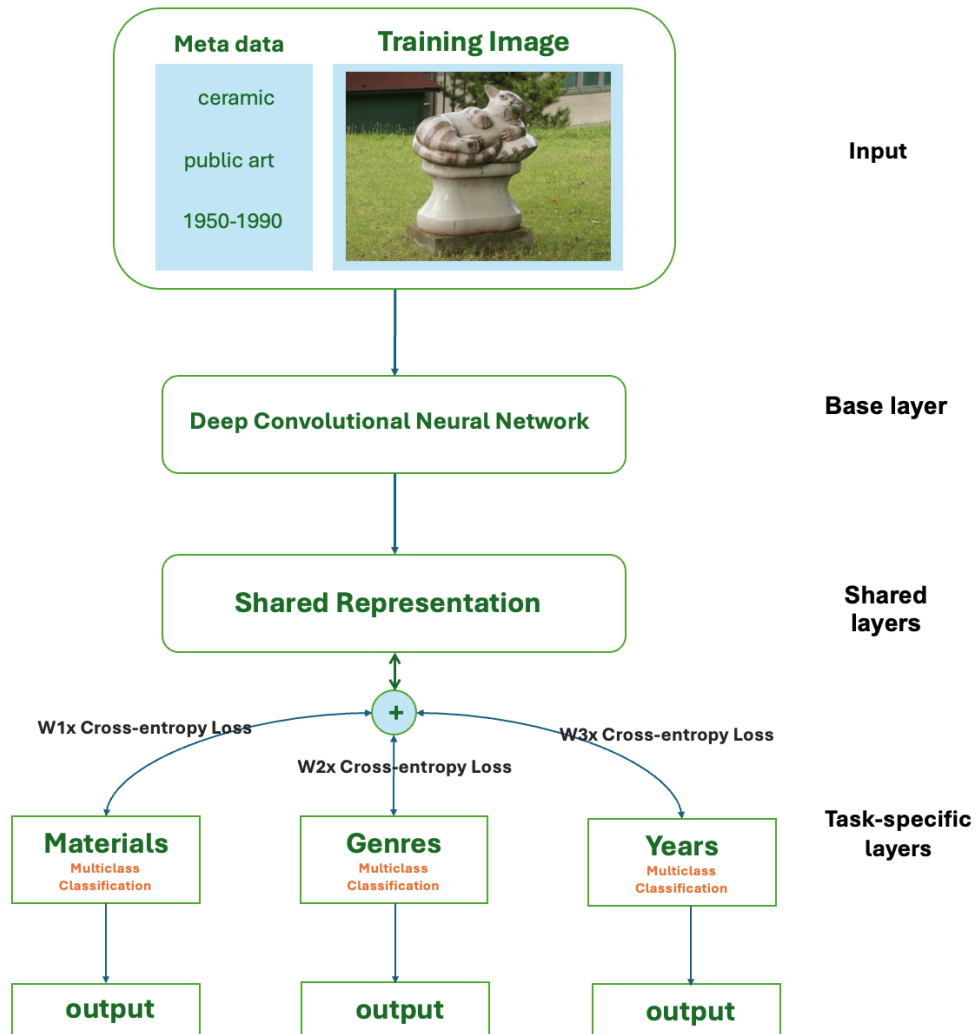


Figure 12. Structure of Multi-task learning in this research

3.2.1 MTL Model structure overview

Figure 12 shows the structure of multi-task learning in this study. Firstly, the Input layer receives the raw data, which consists of an image of the sculpture, as well as the image's corresponding metadata (material, genre and year). Following this, the base layer uses the pre-trained model RestNet50, which extracts high-level features from the input data, such as capturing the shape, texture and colour of the sculpture. The next layer is the shared layer, which is essential for multi-task learning as it allows the model to learn features common to multiple tasks. After the shared layer, the model is split into different branches, each representing a layer dedicated to a different task. Then the losses of the tasks are combined, and the backpropagation of the combined

loss optimises the features of each task in the shared layer. Finally, the output layer returns probability distributions for different classes for each task. The key layers in the multi-task learning model are analysed below.

3.2.2 Base layer

This study uses a neural network model based on the RestNet50 architecture, which has a deep structure of 50 layers. The main highlight of the RestNet network is the Residual Network, which adds the idea of residual learning to the traditional convolutional neural network. As Figure 13 shows, it achieves this by using skip-connections. Skip-connections are extra connections between nodes in different layers of a neural network that skip one or more layers of nonlinear processing (Kovalev and Shishkin, 2020). This solves the problem of gradient dispersion and accuracy degradation (training set) in deep networks, allowing the network to get deeper, ensuring both accuracy and speed control.

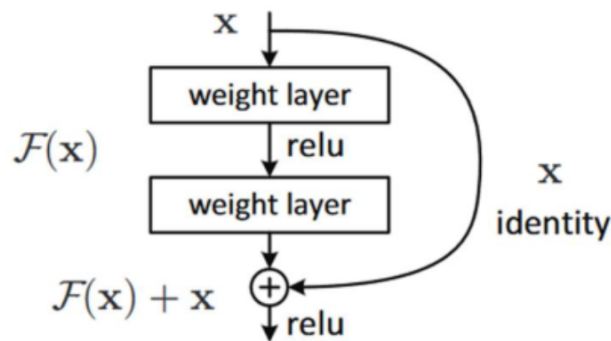


Figure 13. ResNet Skip-connections (Kovalev and Shishkin, 2020)

In this study, RestNet50 is used to extract high-level features from the input image. Specifically, the convolutional output of its last layer was mainly used, and after multiple layers of convolution and pooling, it was able to effectively capture complex features in the image.

3.2.3 Shared layers

The features extracted by the Base layer are subsequently passed to the Shared layer. By integrating the MTAN (Multi-Task Attention Network) architecture (Liu et al., 2019b) with the Shared representation layer, it is ensured that the information transfer

between multiple tasks is coordinated. MTAN uses the Self-Attention layer to adapt to the feature requirements of different tasks. Each task has its specific attention points, and through the attention mechanism, the model can better understand these points of attention and customise them for individual tasks. This dynamic weight assigned approach not only improves the model's ability to generalise across multiple tasks, but also effectively prevents the tasks from interfering with each other.

3.2.4 Task-Specific Layers

Task-Specific Layers optimise the performance of individual tasks by further processing the shared features passed from the previous layer according to the specific needs of each task. This study consists of three main tasks, materials classification, genres classification and years classification. As such there will be three separate Task-Specific Layers. Since all three tasks are classification tasks, the same structure of layers is used. This consists of a linear layer that reduces the dimensionality of the features, then a ReLU activation function to introduce non-linearity, followed by a Dropout layer to prevent overfitting, and finally a linear layer that outputs the scores for the corresponding classes.

3.2.5 Loss Functions and Weighting Strategy

Cross-Entropy Loss is often used to solve multiclass classification problems in deep learning. Since all three classification tasks in this study are multiclass classification, the Cross-Entropy function is used as the loss function:

$$CE = - \sum_i^c t_i \log(f(s)_i)$$

(Kovalev and Shishkin, 2020)

c represents the total number of categories, $c = 9$ in the materials task, $c = 11$ in the genres task, and $c = 13$ in the years task. Each sample in each task has a known and true category label. t_i is the true label for category i . If the current sample is in category i , then $t_i = 1$, and 0 for all other categories. $f(s)_i$ is the predicted probability that the model estimates the sample belongs to category i . This formula is used to

calculate the difference between the probability distribution of the model output and the distribution of the true labels, the smaller the loss indicates that the model's prediction is closer to the true labels.

In this study, the weighting strategy for multitask learning is Dynamic Weight Average (DWA) (Liu et al., 2019b). There are some tricky problems in multi-task learning, such as different tasks have different learning speeds, different tasks have different loss magnitudes, and the dominance of tasks with larger loss may occur. These problems can make the advantage of multi-task learning gradually lost. Therefore, this study uses DWA to dynamically adjust the weights of each task for different training epochs by considering the loss change of each task. With this weighting strategy, tasks that are more difficult to learn will be assigned higher weights.

3.2.6 Implementation Details

The model is built using LibMTL library (Lin and Zhang, 2022), which is based on Pytorch for Multi-Task Learning (MTL). The library provides a variety of multi-task architectures, loss functions and weighting strategies that simplify model developing. The model is trained using the AdamW optimiser with a learning rate of 0.001. Compared to the Adam optimiser, AdamW handles weight decay better and can improve model training. The model applied a weight decay of 0.0005 to prevent overfitting by penalising large weights. The training process was carried out in 30 epochs with a batch size of 10. The parameters of Drop out were found to be best trained at 0.6 after several experiments.

4. Results

This section focuses on showing the training results of the multi-task learning model. This includes performance metrics, loss trends and confusion matrices for each task in multi-task learning. This is then further compared and analysed with the performance metrics of single-task learning.

4.1 Overall Performance

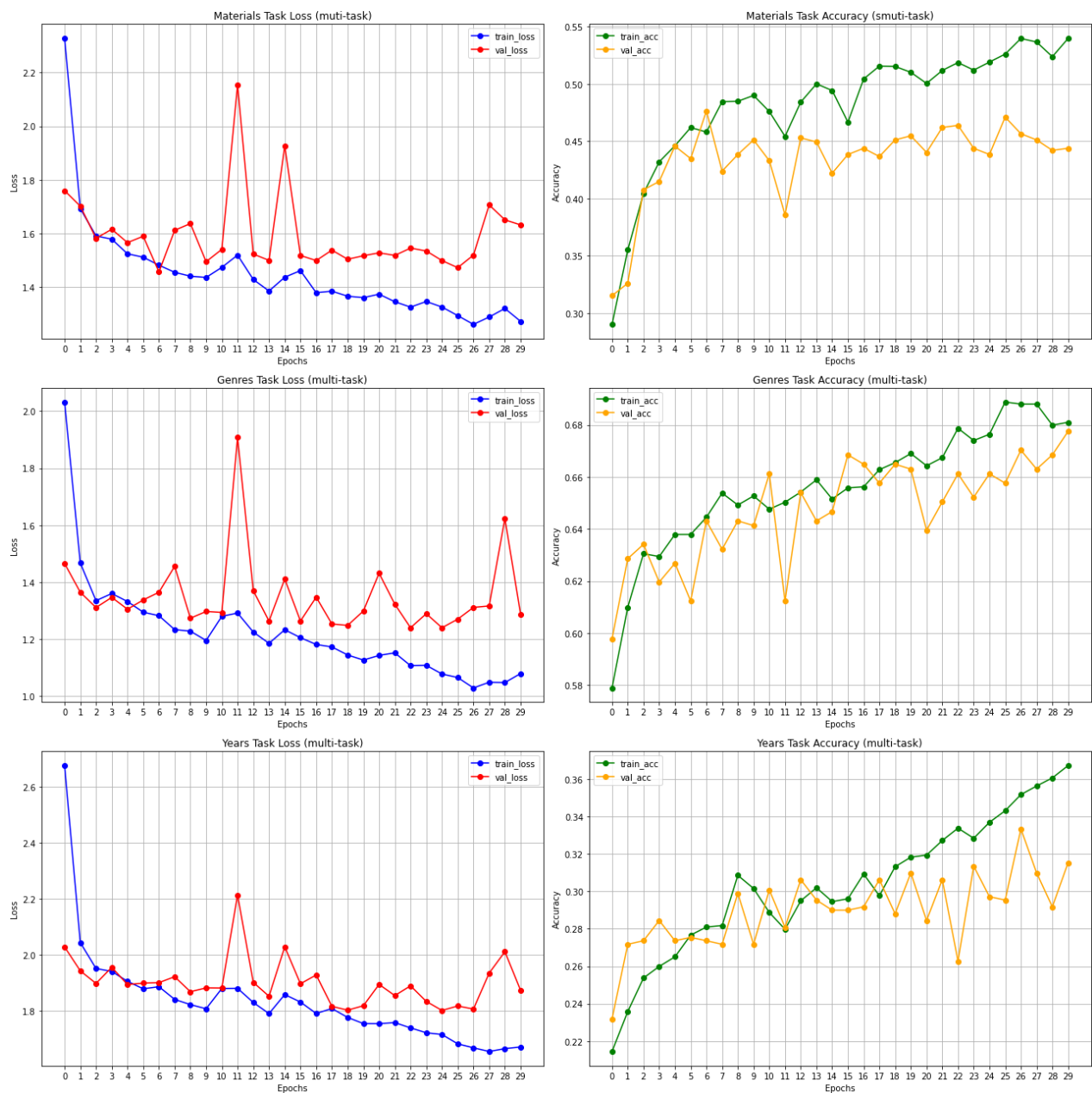


Figure 14. Training and Validation Loss and Accuracy Curves for Materials, Genres, and Years Tasks Using the Multi-Task Learning Model

Figure 14 shows the loss curves and accuracy curves for the three classification tasks of materials, genres, and years for 30 epochs, respectively. Overall, there are varying degrees of overfitting in all three tasks, as the loss in their training set decreases smoothly, while the loss in the validation set fluctuates and decreases at different rates. This is also confirmed in the accuracy curves, which show that there is a difference in accuracy between the training and validation sets, especially in the later stages.

For the materials task (first row), although the validation set loss is decreasing overall, there are significant fluctuations. The accuracy of its validation set fluctuates but has a clear upward trend with increasing epoch. However, in the later stages the accuracy of the validation set lags the training set significantly. For the genres task (second row), the loss is similar to that of the materials task, but the fluctuation in the validation set is slightly milder than that of the materials task. The accuracy of genres task is more consistent between the training and validation sets compared to the materials task, suggesting that this task learns and generalises more efficiently. For the years task (third row), the accuracy on the validation set has more substantial fluctuations than the other two tasks, indicating unstable learning. Taken together, the years classification task is the most challenging task, while the genres classification task has a relatively better performance.

Table 5. Test Set Classification Accuracy (muti-task learning)

Materials Accuracy	Genres Accuracy	Year Accuracy
0.5471	0.6757	0.3062

As shown in the table, the material classification task achieved 54.71% accuracy, the genres classification task achieved 67.67% accuracy, and the years classification task reached 30.62% accuracy. This indicates that this multi-task learning model performed best on genres classification, followed by materials classification, however years classification performed the worst.

4.2 Confusion Matrices for Each Task

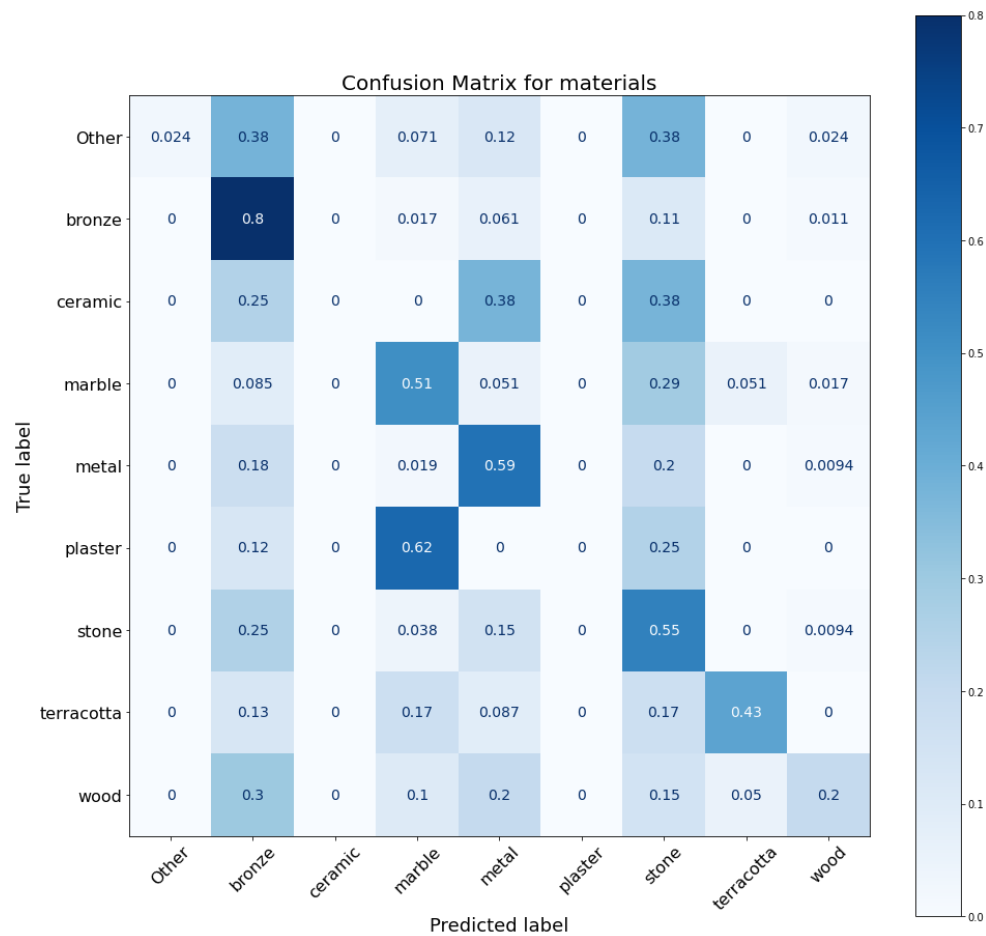


Figure 15. Confusion matrix for materials

Figure 15 shows a confusion matrix map for materials. Diagonal squares indicate correct classifications and off-diagonal squares indicate incorrect classifications. For correct classification, as seen in the diagonal data, the model has a higher accuracy in recognising bronze material with an accuracy of 80%. In addition, the model also has good accuracy in recognising materials such as marble, metal, stone and terracotta, almost all of which have more than 50% accuracy. For misclassification, materials such as ceramic, plaster and wood are significantly misclassified. Firstly 62% of plaster was classified as marble which is a very significant misclassification. Next is ceramic, 25% of ceramic was classified as bronze, 38% as metal and 38% as stone. Finally, wood, which has a more dispersed misclassification, with 30 % classified as bronze, 20 % as metal and 15 % as stone.

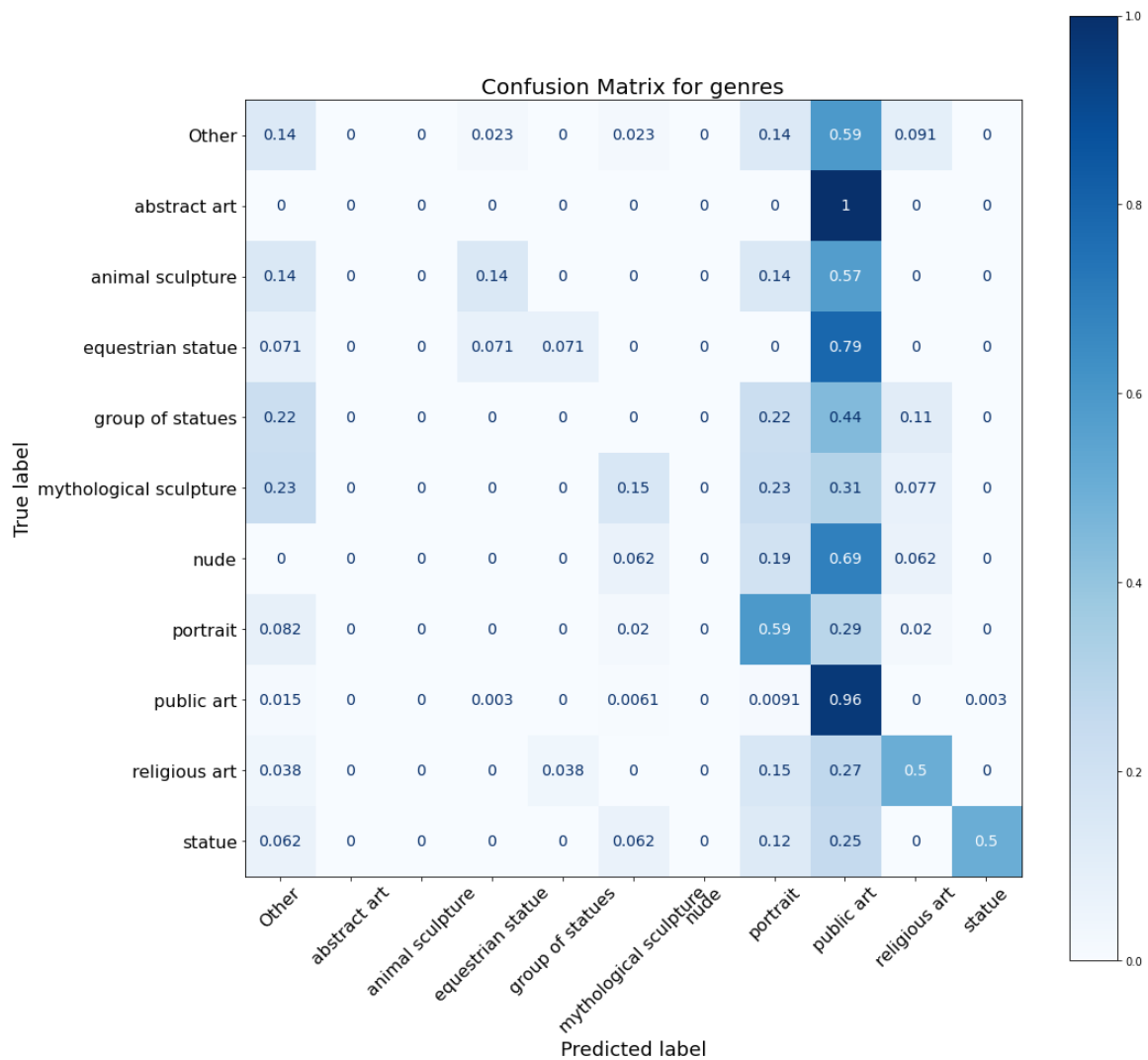


Figure 16. Confusion matrix for genres

Figure 16 shows the confusion matrix map for genres. The figure shows that the model performs very well in classifying public art. 96% of public art is correctly classified, which is the highest classification accuracy of all genres. This is followed by portrait, religious art, and statue, all of which have an accuracy rate of more than 50%. Misclassification is also evident, with almost categories of genres being misclassified as public art to varying degrees. The abstract art misclassification is the most significant, with 100% of all abstract art being misclassified to public art. This is followed by equestrian statue, with 79 % of equestrian statue being misclassified to public art.

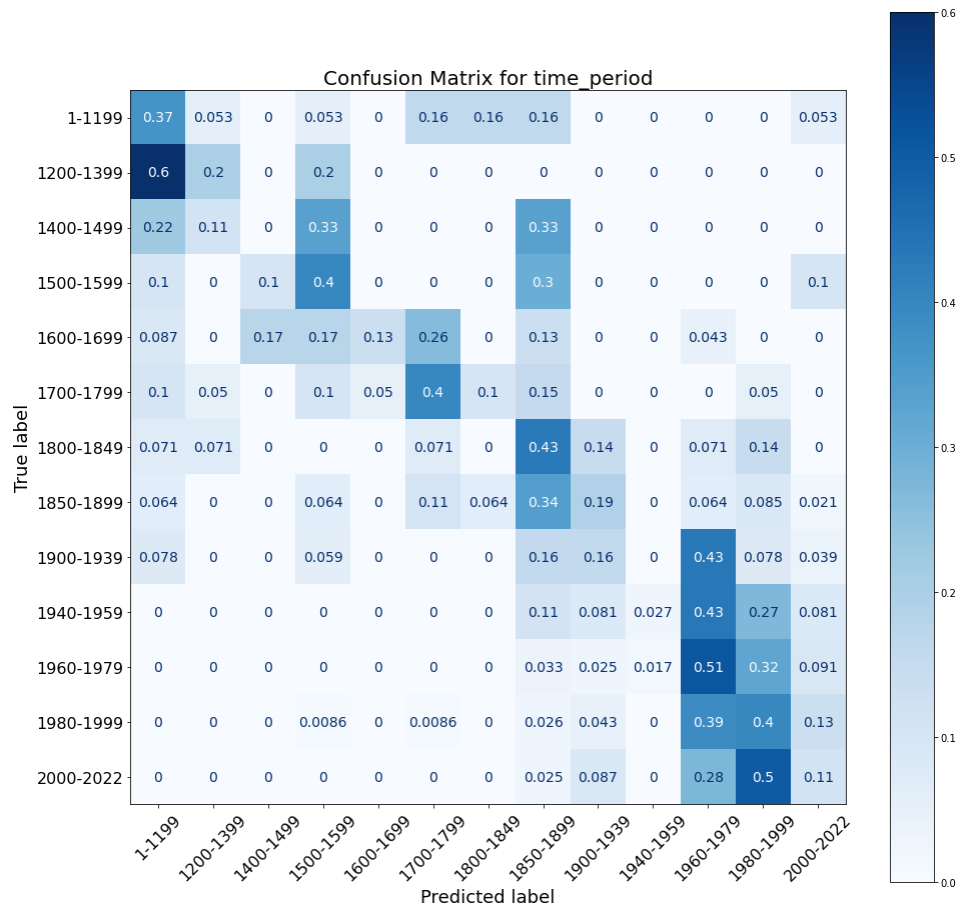


Figure 17. Confusion matrix for years

Figure 17 is a plot of the confusion matrix for years. The matrix shows that the accuracy is highest at 51% in predicting the years 1960-1979. However, most of the misclassifications occur in neighbouring periods. For example, the model misclassifies sculptures from 1200-1399 into the previous period 1-1199; sculptures from 1800-1849 into the next period 1850-1899; and sculptures from 1940-1959 into the next period 1960-1979. This suggests that there are limitations to the model's ability to learn the differences between years.

4.3 Comparison with Single-Task learning

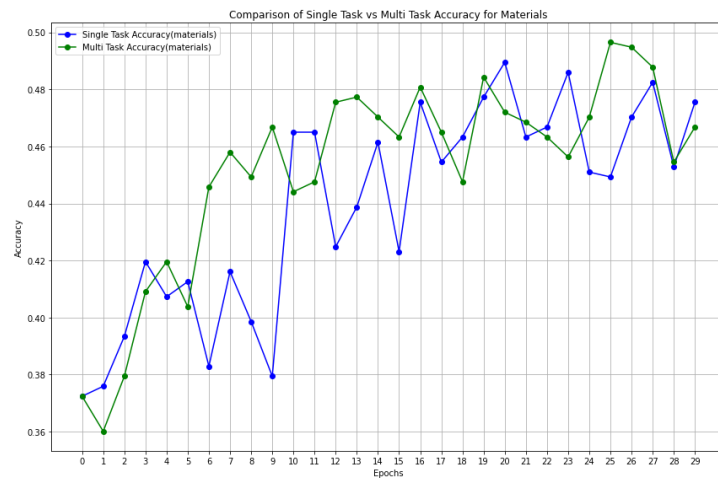


Figure 18. Comparison of single-task learning vs multi -task learning accuracy for materials

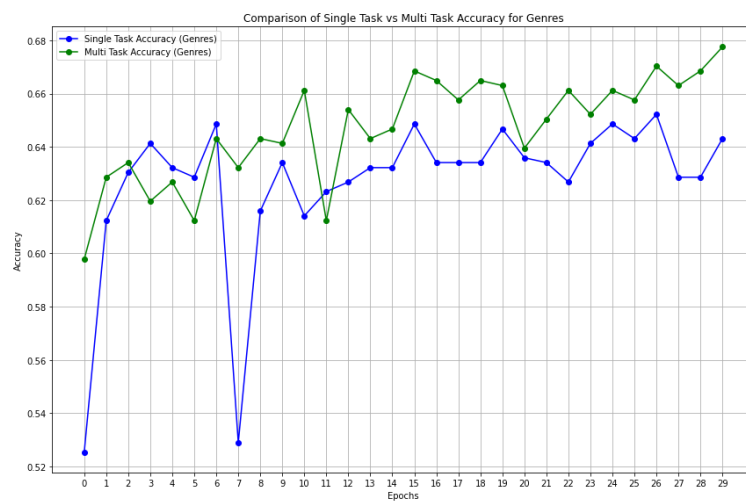


Figure 19. Comparison of single-task learning vs multi -task learning accuracy for genres

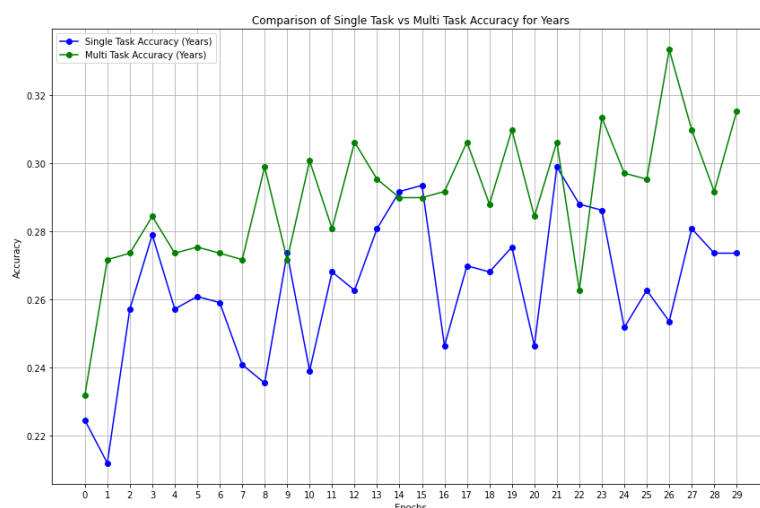


Figure 20. Comparison of single-task learning vs multi -task learning accuracy for years

The above three-line graphs compare the accuracy of materials, genres and years in multi-task learning and single-task learning. Overall, the multi-task learning model outperforms the single-task learning model on all three tasks and most epochs.

In the materials accuracy graph (Figure 18), there are many parts of overlap between multi-task learning model and single-task learning model, but the former has relatively less fluctuation and higher stability. Genres accuracy graph (Figure 19) shows that after the first several epochs the accuracy of Multi-task learning is consistently higher than that of the single-task learning model, and it performs more consistently in the later epochs. The accuracy graph of years (Figure 20) shows that multi-task learning is almost always more accurate than single-task learning, and the accuracy gap between the two models is the largest among the three tasks.

Table 6. Test Set Classification Accuracy

(multi-task learning and single-task learning)

	Materials Accuracy	Genres Accuracy	Years Accuracy
Multi-task learning	0.5471	0.6757	0.3062
Single-task learning	0.5380	0.6413	0.2536

Table 6 summarises and compares the results of the multi-task learning model and the single-task learning model on the test set. Consistent with what is observed in the line graphs, the multi-task learning model shows better performance in all three tasks. The years task has the most significant improvement, boosting accuracy by about 5%. In contrast, the materials task has the least improvement, boosting accuracy by only about 1%.

5. Discussion

This chapter begins by interpreting the results of the model and answering the three research questions. It then summarises the theoretical and practical Implications of this study. Finally, it discusses the limitations of this study and suggestions for future research.

5.1 Interpretation of Results

This section is an in-depth interpretation of results and answers the three research questions in the introduction chapter, which are:

1. How does multi-task learning perform in classifying the materials, genres and years of creation of sculptures? Are there significant differences in performance between these tasks?
2. Did multi-task learning improve the accuracy of the materials, genres and years of creation of sculptures compared to the single-task learning model? In which tasks is the improvement more significant?
3. How do the characteristics of the dataset such as category distribution, and visual features affect the classification of the multitask learning model?

5.1.1 Model performance for multi-task learning

Overall, multi-task learning performs differently on the three classification tasks. It performs well on the genres classification task, achieving about 68% accuracy. The accuracy on this task is more consistent compared to material classification. This suggests that the model is more likely to learn genre-specific features, which may be due to the fact that the differences between genres are more visually apparent. Materials classification task performs average, achieving about 55% accuracy. Although it had some fluctuations in accuracy, the model had a gradual improvement and stabilised at a moderate level of accuracy. This can be attributed to the fact that there is more similarity between the materials. However, the classification task in years was challenging with only about 31% accuracy. This may be because sculptures from

different historical periods have overlapping visual features. Moreover, the period grouping of years in the data preprocessing stage also affects the learning of the model.

The differences in training difficulty and accuracy between tasks in Multi-task learning highlight the differences in its ability to extract features relevant to each task from the shared layer. The higher performance on the genres classification task suggests that the model is more effective at capturing genres-related features in this layer, which is more pronounced than the features of materials and years. In terms of task interaction, although the shared layer can learn features common to all tasks, the features it learns may also interfere with the model's ability to learn a particular task. The next section will analyse whether there is a potential task interference situation by comparing it with single-task learning.

5.1.2 Multi-task learning compared to single-task learning

Multi-task learning showed a significant improvement in accuracy over single-task learning in all three classification tasks. This suggests that the benefits of shared learning provided by multi-task learning outweigh the negative effects of interference between tasks. Although the accuracy of years in single-task learning remains the lowest of the three classification tasks, the improvement is greatest in multi-task learning. It improved by 5% over single-task learning. Obviously, multi-task learning has a greater advantage.

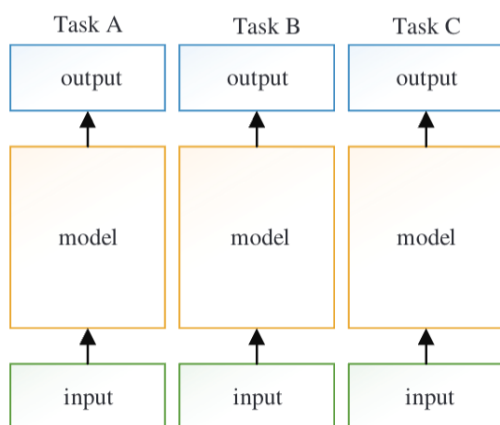


Figure 21. single-task learning network
(Fan, 2023)

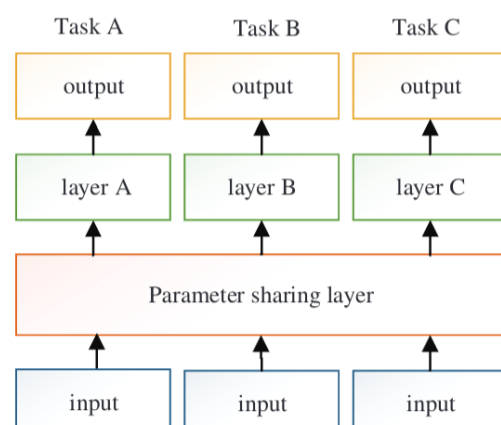


Figure 22. multi-task learning network
(Fan, 2023)

By using multi-task learning, the model can simultaneously learn common features between tasks as well as task-specific features. Figure 21 shows the structure of a single-task learning network, where each task is trained independently and the extraction of features from one task does not affect the other tasks. Figure 22 shows the structure of a multi-task learning network. Different from single-task learning, multi-task learning has a shared layer that shares information between tasks before moving on to task-specific layers. Therefore, if there is a common ground between the materials, genres and years of the sculpture, then using multi-task learning to train the classification model will have a higher accuracy than using single-task learning. The following is an example of the years task, which has the highest accuracy improvement in multi-task learning in this study. We analyse in depth how materials, genres and years share information in the shared layer to improve the accuracy of the years classification task.

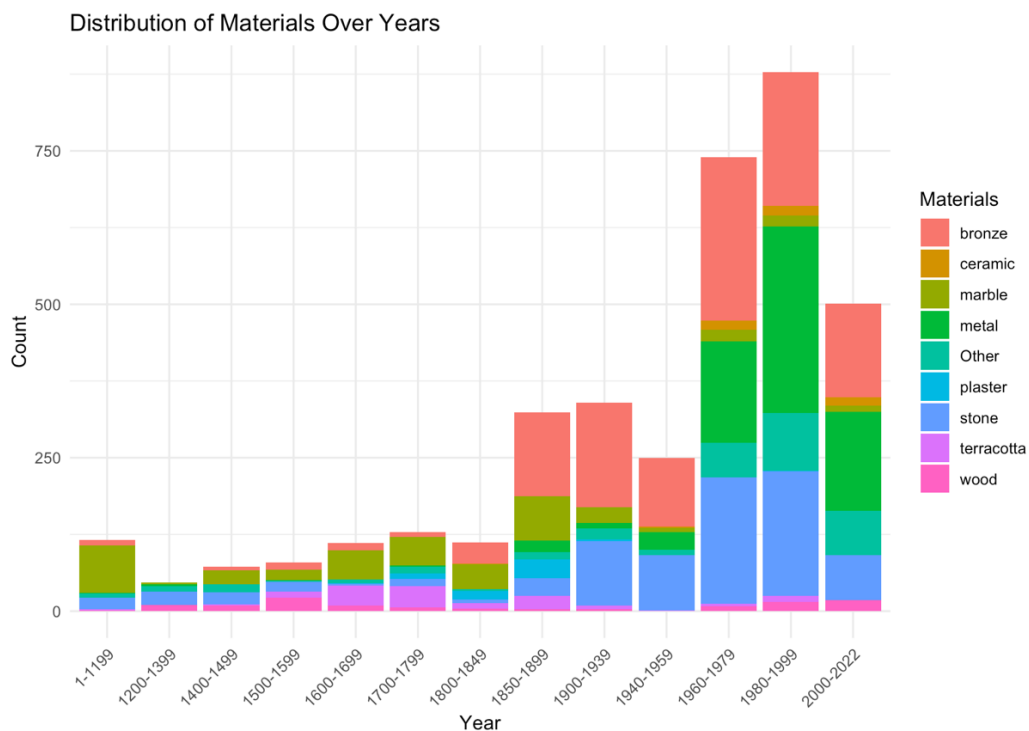


Figure 23. Distribution of materials over years

Figure 23 shows that the frequency of use of different materials in the sculpture is distributed differently across the different periods of the sculpture. Firstly, the heavy use of marble is concentrated in the years 1-1900, where nine years categories are

included. The use of terracotta is concentrated in the years 1600-1799, which will help the model to correctly classify samples of sculptural materials labelled as terracotta into this period. After 1850, there is a significant increase in the use of bronze in sculpture. After 1960, metal and plaster are more heavily represented, suggesting that these two materials began to be used more extensively in sculpture during this period. Wood, on the other hand, was used evenly in sculpture in almost every period, suggesting that the information on material is not very helpful in classifying years.

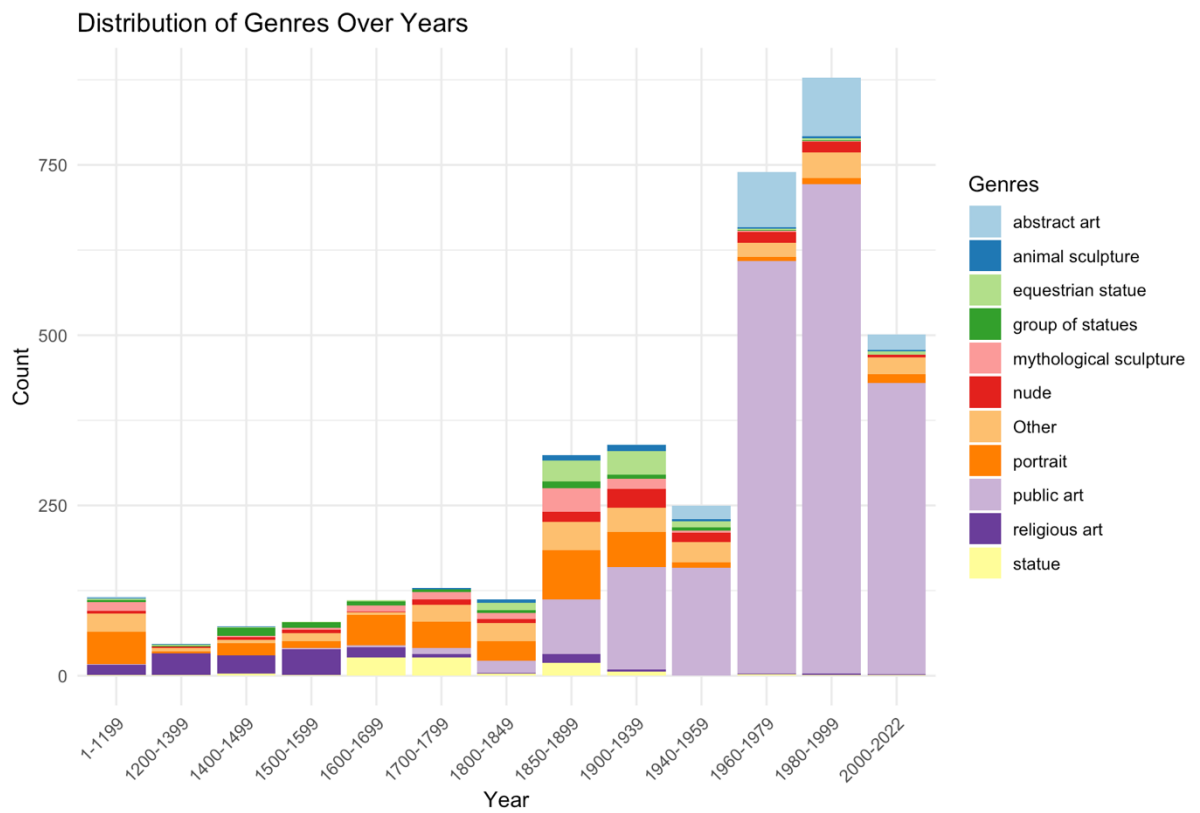


Figure 24. Distribution of genres over years

Similarly, Figure 24 shows that the percentage of genres in sculpture varies across years. Religious art is mainly concentrated in the period 1-1600, statue is mainly concentrated in the period 1600-1899. After 1960, the type of sculpture becomes dominated by public art and abstract art. The above relationship between genres and years is beneficial for the multi-task learning model to improve the learning of years.

However, portrait has a certain degree of proportion in all periods. This suggests that the genres of portrait run through ancient and modern times, and the relationship between it and years may not be large enough to help better classify years.

In summary, the materials task and the genres task in multi-task learning provide contextual information that is beneficial to the years classification task. Their contributions in the shared layer ultimately improve the accuracy of the classification of years.

5.1.3 Effects of dataset characteristics on model classification

Firstly, the effect of data category imbalance on model classification. In the materials task, it is observed that bronze has the highest rate of correct classification and bronze is one of the categories that has the highest number of samples under the materials variable. Similarly marble, metal, stone and terracotta have more samples, which helps the model to learn better and makes these categories to be classified correctly to some extent. However, for materials categories such as ceramic, plaster and wood which are often confused by the model, they are categories with very small sample sizes. Such categories may be under-represented, leading to a greater tendency for the model to misclassify. In the genres task, the problem of imbalanced category distribution is even more severe than in the materials task. the sample size of the public art category far exceeds the sample size of the other categories, which obviously biases the model towards learning the features of public art. However, this also leads to another problem, where almost all genres categories are partially misclassified into public art. This suggests that public art is over-represented during the training process of the genres task. When the model encounters other uncertain genres, it is more biased towards predicting the majority class.

Secondly, the effect of image visual feature similarity on model classification. Some categories of visual features may have too similar attributes, which leads to the model not being able to adequately differentiate between certain materials, genres or years. For example, in the materials classification task, 62% of plasters were incorrectly classified as marble. As can be seen in Figure 25, plaster and marble are visually very

similar, both being white and having a certain glossiness. Common sense would suggest that marble would be glossier than plaster. However, due to the difference in lighting and the scene in which the sculpture images were taken, this results in the subtle differences in material texture not being present in the images. This ultimately makes it difficult for the model to determine the category from the image.



Figure 25. Examples of Plaster and Marble Sculptures

In the genres classification task, 100% of the abstract art was misclassified as public art and 79% of the equestrian statue was misclassified as public art. Figure 26 shows that abstract art and public art have similar visual characteristics, such as minimalism, outdoor and large-scale installations, and so on. Abstract art and public art also have similar attributes, such as outdoor environments, human sculptures, and so on. Furthermore, the category of public art is a very wide category and any sculpture that is in a public space belongs to this category, which leads the model to learn the characteristics of other genres in this category as well.



Figure 26. Examples of public art and equestrian statue

Finally, the effect of temporal data on model classification. One feature that can be observed in the confusion matrix plot for years is that misclassification almost always occurs in neighbouring time periods. The model confuses the correct period in which the sculpture was made with its neighbouring time periods. For example, in the confusion matrix there are 32% of sculptures from 1960-1979 that are misclassified as 1980-1999. The comparison plot of these two years labels (Figure 27) reveals that the materials and genres in both periods are so extensive that visually the sculptures from these two periods are extremely similar. Theoretically, sculptures from different periods have different characteristics and they evolve over time. In practice, on the one hand, there is a transition period in the evolution of sculpture types, rather than a sudden change from one period to another. This is because artists are often influenced by the work of their predecessors, and their sculptural characteristics will have coherence and similarity between neighbouring periods. This leads to the fact that sculptures from neighbouring periods may share many visual characteristics. On the other hand, certain materials and carving techniques may have been used for more

than one period, making it difficult for the model to correctly assign the sculpture to the right period.

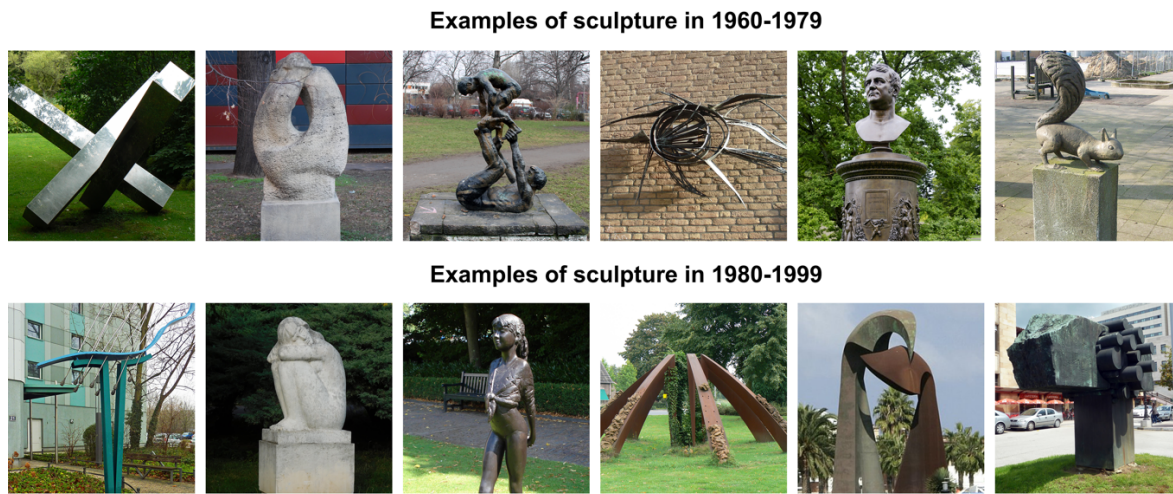


Figure 27. Examples of sculpture in 1960-1970 and 1980-1999

5.2 Implications of the Study

In terms of theoretical Implications, this study expands the research on multitask learning in the field of art and materials. Firstly, there is a longitudinal expansion. As mentioned in the literature review, in the field of art categorisation, the research base of multi-task learning has either focused on paintings, classifying and predicting artists, genres, and periods in paintings, or based on the broader context of art, classifying paintings, architecture, and sculpture. Yet few studies have used multitasking learning to specifically focus on sculpture as an art form. Secondly, in terms of horizontal expansion, most studies in the field of material classification are still using basic deep learning models, i.e., single-task learning models, to perform classification. This study fills in the gaps in these two areas by using multi-task learning to classify materials, genres, and years of sculpture. Although the accuracy of the model is not outstanding, it can still provide a foundation for other researchers to further study multi-task learning in the sculpture or material fields.

In terms of practical Implications, first, the model can benefit museums, galleries and other cultural institutions. Through the model, it is possible to accurately categorise

sculptures in terms of material, genre and year of creation at the same time. This not only greatly improves the classification efficiency, but also saves labour costs and helps curators to manage large collections more efficiently. Secondly, integrating the model or its algorithms into digital art platforms can help users discover and filter sculptures more efficiently and provide more accurate recommendations to optimise the user experience.

5.3 Limitations and Recommendations

The first is the limitation of the dataset. The data sample for this study was 3679. This size of dataset is clearly insufficient due to the large number of categories for all three categorisation tasks. It also leads to an uneven distribution of categories for all three variables. This makes the model unable to learn the features of each category in each task well during the training process. In general, oversampling is a good way to balance the category imbalance, but since we are training three tasks at the same time, balancing the category imbalance in one task may lead to an even more imbalanced category distribution in the other two tasks. Therefore, in future research, for multi-task learning models may need more detailed methods to deal with data imbalance. For example, using GAN to generate synthetic samples for categories that are underrepresented in all three variables in the dataset to improve category imbalance.

The second is Methodological Limitations. the multi-category labelling of materials and genres in the raw data was simplified to single-category labelling during the data processing stage. This may be an important factor affecting model learning. The reason is that several sculptures are made with composite materials or belong to multiple genres at the same time. In addition to simplifying the labels, the categories of materials and genres were also simplified. Since they have hundreds of categories, it makes sense to simplify them by clustering and merging, but this results in the categories losing a large amount of detail. This process also introduces variance, leading to confounding when the model predicts. Therefore, in future studies, researchers can try to retain multiple labels in the categorical variables and perform multi-label categorisation in multi-task learning. On the other hand, try to retain as

many categories as possible when there is enough data, which can make the boundaries between categories clearer.

Finally, the limitation of Backbone. The backbone used in this study is Restnet50, which was released in 2015. Nowadays there have been many studies based on this backbone. However, Restnet50 is still slightly inferior to a more powerful big data model DINOv2 (Oquab et al., 2024) which was released in 2023. With powerful self-supervised learning and more efficient feature extraction, DINOv2 can be widely used in various computer vision tasks. However, due to limited computational resources, DINOv2 was not used in this study. Future studies could use DINOv2 as a backbone and compare its performance with that of Restnet50.

6. Conclusion

In this paper, a deep neural network model for multi-task learning is developed based on LibMTL library for classifying sculpture images. The model consists of three main tasks, which are materials classification, genres classification and years classification of sculptures, which correspond to 9, 11 and 13 categories, respectively. The model utilises a pre-trained model ResNet50 for feature extraction. Various methods are used to improve the classification quality, such as adjusting the parameters of the optimizer, adding drop out layer and weight decay. The model finally achieved 68% accuracy for genres classification, 55% for materials classification and 31% for years classification. Then the results of multi-task learning were compared with those of single-task learning. It was found that multi-task showed significant advantages on all three tasks. By analysing the confusion matrices for the three tasks in multitask learning, it was concluded that the features of the sculpture dataset have a significant impact on the model to correctly classify the labels. In real-world applications, the model can be applied to art institutions such as museums, enabling sculptures to be effectively managed; or integrated into digital art platforms, thus optimising user experience. However, this study has some limitations in terms of the dataset, methodology and backbone used. This points the way and lays the foundation for multitask learning research in the sculpture and material fields in the future.

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Appendix

Final Dataset:

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/final%20dataset%20of%20sculpture.csv>

Data Pre-processing code:

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/data%20preprocessing.R>

Multi-task learning Modeling code:

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/Resnet50-MTL-model.ipynb>

Single-task learning Modeling code:

1. Genres Classification

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/Resnet50-STL-genres.ipynb>

2.Years Classification

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/Resnet50-STL-years.ipynb>

3.Material Classification

<https://github.com/Aastra01/MTL-for-sculpture-classification/blob/main/Resnet50-STL-materials.ipynb>